MAINCRAFTS TECHNOLOGY
www.maincrafts.com
hr@maincrafts.com

Data Analytics & Business Intelligence - Task 2

Advanced Data Analysis, SQL Joins & Business KPI Dashboard

Objective

In **Task-1**, you learned how to:

- Clean raw data
- Perform basic Excel analysis
- Write simple SQL queries (GROUP BY, SUM, COUNT)
- Create a basic dashboard
- Extract business insights

In **Task-2**, you will move to the **next practical level** by learning how to:

- Work with multiple related datasets
- Use SQL JOIN operations
- Create calculated business metrics
- Perform comparative analysis
- Build a professional KPI-based dashboard

This task focuses on **structured business reporting**, not predictive modeling.

Why This Task Matters (Industry Context)

In real organizations:

- Data is rarely in one table
- Analysts combine multiple datasets
- KPIs are calculated, not just extracted
- Management dashboards compare performance across time and categories

Task-2 introduces interns to **real business intelligence workflows**.

What You Will Build

You will work with **two related datasets**, for example:

1. Orders Table
2. Customers Table

(Optional: Products table)

You will:

- Combine datasets using SQL JOIN
- Calculate KPIs
- Perform trend and performance analysis
- Create a structured management dashboard

Dataset Structure (Example)

Orders Table

- Order_ID
- Order_Date
- Customer_ID
- Product_Category
- Sales
- Quantity
- Profit
- Discount

Customers Table

- Customer_ID
- Customer_Name
- Region
- Segment

Part A: SQL Analysis (Moderate Level)

Step 1: Import Both Tables into Database

Create tables in MySQL / SQLite.

Step 2: Perform INNER JOIN

Combine orders and customers:

```sql
SELECT
    o.Order_ID,
    o.Order_Date,
    c.Customer_Name,
    c.Region,
    o.Product_Category,
    o.Sales,
    o.Profit
FROM Orders o
INNER JOIN Customers c
ON o.Customer_ID = c.Customer_ID;
```

This teaches real-world relational database usage.

Step 3: Calculate Key KPIs Using SQL

1. Total Sales by Region

```sql
SELECT c.Region,
       SUM(o.Sales) AS Total_Sales
FROM Orders o
JOIN Customers c
ON o.Customer_ID = c.Customer_ID
GROUP BY c.Region;
```

---

2. Profit Margin by Category

```sql
SELECT Product_Category,
       SUM(Profit)/SUM(Sales) AS Profit_Margin
FROM Orders
GROUP BY Product_Category;
```

3. Monthly Sales Trend

```sql
SELECT
    MONTH(Order_Date) AS Month,
    SUM(Sales) AS Monthly_Sales
FROM Orders
GROUP BY Month
ORDER BY Month;
```

---

4. Top 5 Customers by Revenue

```sql
SELECT c.Customer_Name,
       SUM(o.Sales) AS Total_Revenue
FROM Orders o
JOIN Customers c
ON o.Customer_ID = c.Customer_ID
GROUP BY c.Customer_Name
ORDER BY Total_Revenue DESC
LIMIT 5;
```

Part B: Excel Advanced Analysis

Step 4: Use Pivot Tables for Comparison

Create:

- Sales by Region and Segment
- Profit by Category and Region
- Monthly Sales Trend
- Top 10 Customers

Step 5: Add Calculated Columns

In Excel:

- Profit Margin = Profit / Sales
- Average Order Value = Total Sales / Total Orders

Use formulas:

- IF
- SUMIFS
- COUNTIFS

Part C: Power BI / Tableau KPI Dashboard

Step 6: Import Cleaned Dataset into Power BI / Tableau

Create a structured dashboard including:

KPI Cards

- Total Sales
- Total Profit
- Profit Margin
- Total Customers

Visualizations

- Bar Chart: Sales by Region
- Column Chart: Profit by Category
- Line Chart: Monthly Sales Trend
- Table: Top Customers
- Pie Chart: Segment Distribution

Step 7: Add Filters / Slicers

Add filters for:

- Region
- Product Category
- Date
- Customer Segment

Make dashboard interactive.

Step 8: Business Performance Analysis

Answer these business questions:

1. Which region generates highest revenue?
2. Which category has highest profit margin?
3. Are discounts reducing profitability?
4. Which customer segment is most valuable?
5. Is there a seasonal sales trend?

Expected Output

At the end of Task-2, you must have:

- SQL JOIN queries executed
- KPI calculations
- Comparative analysis
- Structured BI dashboard
- Business summary report

Business Summary Report (Mandatory)

Prepare a 2–3 page report including:

- KPI Overview
- Regional Performance
- Category Performance
- Customer Analysis
- Recommendations for business growth

Key Learning Outcomes

After Task-2, interns will be able to:

- Work with relational databases
- Perform SQL JOIN operations
- Calculate business KPIs
- Build structured BI dashboards
- Interpret multi-dimensional business performance

Deliverables

Mandatory:

1. SQL query file (.sql)
2. Excel file with pivot analysis
3. Power BI (.pbix) or Tableau dashboard
4. Dashboard screenshots
5. Business performance report (PDF)

Optional:

- Data model diagram
- Advanced pivot formulas

1. SQL (JOINs, Aggregations, Business Queries)

SQL JOIN Explained Clearly

https://www.youtube.com/watch?v=9yeOJ0ZMUYw

SQL Joins Full Explanation (INNER, LEFT, RIGHT)

https://www.youtube.com/watch?v=0OQJDd3QqQM

GROUP BY & Aggregations in SQL

https://www.youtube.com/watch?v=9Pzj7Aj25lw

SQL for Data Analysis (Practical Business Queries)

https://www.youtube.com/watch?v=7S_tz1z_5bA

SQL Subqueries & Filtering (Moderate Level)

https://www.youtube.com/watch?v=2GZXoGqS3D4

Free SQL Practice Platforms

SQL Practice Website

https://www.sql-practice.com/

SQLite Online Editor

https://sqliteonline.com/

MySQL Installation Guide

https://dev.mysql.com/doc/

---

2. Excel (Pivot Tables, KPI Calculations, Advanced Analysis)

Pivot Tables in Depth

https://www.youtube.com/watch?v=UsdedFoTA68

Excel SUMIFS & COUNTIFS

https://www.youtube.com/watch?v=ShBTJrdioLo

Excel Dashboard Tutorial (Business Focused)

https://www.youtube.com/watch?v=K74_FNnlIF8

Calculating Profit Margin & KPIs in Excel

https://www.youtube.com/watch?v=4mKp8JqkXfQ

Official Microsoft Excel Help

https://support.microsoft.com/excel

---

3. Power BI (KPI Dashboards & Data Modeling)

Power BI Desktop (Free Download)

https://powerbi.microsoft.com/desktop/

Power BI Beginner to Intermediate Course

https://www.youtube.com/watch?v=AGrl-H87pRU

Power BI Dashboard Design Tutorial

https://www.youtube.com/watch?v=3oC3m-2Vb1k

Power BI Data Modeling Basics

https://www.youtube.com/watch?v=0a2oHbeSx8E

Creating KPI Cards in Power BI

https://www.youtube.com/watch?v=Z2zWf9v1nZQ

Official Power BI Documentation
https://learn.microsoft.com/power-bi/

---

4. Tableau (Free Alternative to Power BI)
Tableau Public (Free)
https://public.tableau.com/
Tableau Dashboard Beginner Tutorial
https://www.youtube.com/watch?v=2P9hE0XjQ2Q
Tableau Joins & Data Modeling
https://www.youtube.com/watch?v=7QqKjQ6nFJY
Official Tableau Training
https://www.tableau.com/learn/training

---

5. KPI & Business Insight Development
Understanding Business KPIs
https://www.youtube.com/watch?v=Qd8z8QYp4xE
Writing Data Insights Professionally
https://www.youtube.com/watch?v=7zQ6cM8yQmE
Dashboard Best Practices
https://www.youtube.com/watch?v=V6LQpYxkF8U

---

6. Sample Datasets (Free)
Kaggle Datasets (Search: Superstore, Sales Data)
https://www.kaggle.com/datasets
Google Dataset Search
https://datasetsearch.research.google.com/
Data.gov
https://www.data.gov/

---

7. End-to-End BI Project Examples
SQL + Excel + Power BI Project
https://www.youtube.com/watch?v=qfyynHBFOsM
Sales Dashboard End-to-End Project
https://www.youtube.com/watch?v=GfXFmVfSg5M
Business Intelligence Project Walkthrough
https://www.youtube.com/watch?v=Z2zWf9v1nZQ

---

8. Community Support
Stack Overflow (SQL + Excel + Power BI)
https://stackoverflow.com/
Power BI Community
https://community.powerbi.com/
Tableau Community
https://community.tableau.com/

Reddit Data Analysis
https://www.reddit.com/r/dataanalysis/

---

9. Recommended Learning Plan for Task-2

Day 1
SQL JOIN and aggregation practice
Day 2
Excel pivot tables and KPI calculations
Day 3
Power BI data import and modeling
Day 4
Dashboard creation with filters
Day 5
Insight writing and documentation

---

10. Essentials Checklist for Task-2

You must clearly understand:

- How SQL JOIN works
- How to calculate KPIs (Profit Margin, AOV, etc.)
- How to build multi-table analysis
- How to design a clean dashboard
- How to interpret business performance
- How to write structured business recommendations